

Third addendum to the 3.0.9/3.1.1 SOLPS-ITER Release Notes

August 6th, 2024

This update to the *3.0.9/master* and *release/3.1.1* code branches is meant to apply, for the most part, a critical bug fix relevant to OpenMP runs. Indeed, it was found that, when running the code with a `gfortran` compiler, OpenMP runs would lead to segmentation faults and/or produce garbage results. Fixing that problem unveiled several issues with the code's OpenMP implementation, which have all now been fixed. I want here to extend my sincere gratitude to [Saxena Gaurav EXT](#) from the BSC ACH, without whose expert assistance I would have been unable to resolve this issue.

There are many other changes introduced with this update, also listed below, several of which are preparatory steps related to the porting of the code at ITER to new Red Hat 9 architecture nodes (more on this in a separate notice). To obtain the new 3.0.9 *master* version, use the *solps-iter_update* script. Be mindful that, because many of the setup and configuration files change with this update, the code will most likely fail to compile the first time around. In that case, close your session, restart it to load the updated setup and configuration files, and attempt the compilation again. Same story for the *release/3.1.1* branch, but using the *solps-iter_update_3.1.1_release* script, of course.

Here is a rundown of the various changes made to the code with this update:

- To obtain correct OpenMP run results, the code has been restructured so that all matrix solver arrays are initialized in parallel during the code set-up phase, and then are passed as `COPYIN` variables in the actual solution loops. Also, the solvers are restarted on every call when running inside a parallel loop. All solver array variables have been renamed with unique names to avoid clashes when the same thread has to handle more than one solver method. The `last_solve_X` arrays have been given an additional species dimension ($0:ns, 0:nreg(0)$) for the same purpose, allowing them to become `SHARED` variables for the OpenMP solution loops. The species index *ns* relates to equations which do not depend on the species number (total momentum, potential, and energy equations).
- The `OMP_STACKSIZE` environment variable was added to the set of variables defined via the *set_openmp* script, as it is needed by the `gfortran` compiler, while the Intel compiler can use the `KMP_STACKSIZE` variable instead. Default values for both have been set to 128 MB (or 256 MB if using *set_openmp*).
- All the `!$OMP` sentinels have been un-indented to start in column 1 of the source files, which was not always the case. The code also now makes systematic use of the `_OPENMP` pre-processor pragma in the **B2.5** source code to avoid warnings from compilers not expecting OpenMP commands.
- The *roxa* mass density array is made local in the *b2sifr* routine to allow for better vectorization of the loops computing the friction terms.
- The parallel computation of *b2stbr_phys_sna* has been corrected.
- The code now links to **Eirene** MsV version 1.0.8, which includes a new scheme for obtaining angular-resolved spectra of incident particles onto walls, implemented by Henri Kumpulainen, and may be of interest when computing post-facto sputtering rates. This capability is turned on by means of a new `ISPLDEG` variable in block 10F of the **Eirene** input file. See the **Eirene** manual for details.
- A new `adas_extrap` switch is introduced, following a report by [Senichenkov Ilya EXT](#), to avoid spurious extrapolations of ADAS tabulated data. This switch is meant to be used in *b2ar.dat*, but can also be included in *b2mn.dat*, for the purpose of plotting the ADAS rates using **b2plot**. The default value (`adas_extrap = 1`) provides an extrapolation that grows no faster than the 3/2-th power of the variables along which the extrapolation is taking place. The setting `adas_extrap = 0` recovers the old behavior of an unrestricted log-log extrapolation, and `adas_extrap = 2` asks for no extrapolation at all. Users that suspect their atomic rates may be suffering from bad extrapolations should rebuild their *b2frates* files with the new code.
- Thanks to [Horsten Niels EXT](#), a discrepancy was discovered in the *fort.44* file concerning the `sr_cml` (molecule dissociation rate) and `edissm1` (energy deposited by the Franck-Condon atoms into the atom fluid) arrays computed by **Eirene**. These two arrays had the wrong units and were not divided by the cell volume as they should have been. This is now corrected. A *fort.44* file with the header 20240627 will contain the corrected data. If the code reads an earlier file, it will convert the data on the fly. This conversion can also be done by means of the **b2yt** program. When saving the run data into an MDSplus database entry with **b2md**, the database version number has been incremented to 31 to indicate that these arrays now contain the corrected values.
- Following suggestions from [Voskoboinikov Sergey EXT](#), `SAVE` attributes and initializations have been added to various quantities involved in the Zhdanov closure scheme. The diagnostic output filenames from the *b2tra1* routine have also been streamlined.
- New IPCCR configuration files and the associated domain definition have been brought in (thanks to [Hecko Jan EXT](#) for providing those). Domain definitions for the NSCC and DLUT clusters were also added.
- Thanks to [Coster David EXT](#), we now have the brand new set of ionization potentials as per the latest NIST database update, and the code examples have been rebuilt using these new values, although no significant differences are found in the results.
- Support for the Intel `ifort` 2021, `ifx` 2023, and `GCC-13.2.0` compilers is now available. Some code clean-up was performed to obtain a compilation as warning-less as possible with these new compilers.
- Compilation of **B2.5** on MARCONI with the Intel compiler now adds the `-fp-model=precise` flag to avoid an error when testing grid data in **b2ag**. Thanks to [Westerhof Egbert EXT](#) for the bug report and testing on this one. Therefore, Intel MARCONI users should recompile their **b25** and **b25eirene** targets from scratch to apply this flag.
- As requested by [Piras Anthony EXT](#), the data in the *output/* diagnostic files is now always written with three exponent digits.
- The compilation of the `nc_reduce` utility program is streamlined so it always uses the serial object files. If you only compile a parallelized version of SOLPS-ITER, you may need to delete that executable first before relaunching the compilation. For people that have private configuration files, this requires the definition of a new `OBNDDEST` variable in your `$SOLPSTOP/modules/B2.5/config/config.$HOST_NAME.$COMPILER(.local)` file, which points to `OBNDIR` instead of `OBJDIR` like `OBJDEST` does. See its usage in any of the distributed configuration files. If you are using the `gfortran` or `pgf90` compiler, this is already taken care of for you in the corresponding *config.common.\$COMPILER* file.
- Private configuration file owners should be aware that the *Makefile* expects environment variables for the serial FC Fortran compiler and the MPI compiler `MPI_FC` to be declared inside the *SETUP/config.\$HOST_NAME.\$COMPILER(.local)* file, if you are not using the system defaults. For everybody else, this is already taken care of.
- A bug fix was introduced to allow the command `10 wlld` to run from a **B2.5** standalone run, using a coupled **b2plot** executable, even though it will produce no useful wall loading data (since **B2.5** standalone is not aware of the true wall geometry).
- A bug fix was put in to avoid a division by zero error when running the code compiled with the `-DASTRA` option but not actually coupled with **ASTRA**.
- [Reiter Detlev EXT](#) provided a bug fix to the **Eirene** printout of the sheath potential drop at the targets (activated with the `TRCINT` switch) so that, if **Eirene** is using the sheath potential drop computed by **B2.5** (`eirene_sheath_pot=1`, the default behavior), this is what is printed out, instead of the potential drop **Eirene** would compute internally.

- A bug fix was provided by [Westerhof Egbert EXT](#) related to the reading of the reference pressure cell number for the pressure feedback scheme in **Eirene**.
- The error message given by **Eirene** when the wall energy variable EWALL is set to zero, which is not supported in the SOLPS version of **Eirene**, has been made more clear. This value can be set inside the **DivGeo** model as the `Wall Temperature` field of the `General Surface Data` dialog window.
- To allow for the proper treatment of pre-processor pragmas when compiling **Eirene**, some files were renamed from `.f` to a `.F` extension, the latter being understood by the *Makefile* to require pre-processing.
- When doing a first installation of the code, it may be necessary to do a `gmake depend` as part of the initial compilation, so the **Eirene** CMake can properly find the `b2mod_dimensions` file. The download and installation instructions document has been modified to indicate this. A note was also added to reflect the fact that the SSH key that the ITER Git server expects should be 1024 bits.
- A few typos related to compilation with the GGD library have been fixed, and some safeties added in the code when compiling in presence of an IMAS module, but no associated GGD one, in which case the code will compile, but the *Makefile* will issue a warning that the build may be incomplete.
- The *runs/Makefiles* now take into account the `.openmp` extension for grabbing the **Eirene** executable, when applicable (thanks to [Hecko Jan EXT](#) for that bug fix).
- The dependency on the Sonnet library for the main code compilation has been removed. That library is now only necessary if one compiles the conversion programs to read older SOLPS4.x results files (**fxdr** and **b2sxdr**). A new `x86_64_rhel9/` directory has been added for building the Sonnet library for the Red Hat 9 architecture.
- The code documentation has been updated to reflect the changes described above. The Manual also now contains a fuller description of the available options for setting up the transport coefficients for the various equations being solved. Some other mistakes in the inline code descriptions have also been corrected following user reports.